MISC

Quadratic formula:

$$ax^2+bx+c=0 \Rightarrow x_{1,2}=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

Monotonically in-/decreasing:

$$\forall x.\ (f'(x)>0) \Rightarrow a>b \Leftrightarrow f(a)>f(b)$$
$$\forall x.\ (f'(x)<0) \Rightarrow a>b \Leftrightarrow f(a)<f(b)$$

RSS:

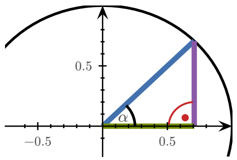
$$\mathrm{RSS}=\sum_i (y_i-f(x_i))^2$$

TRIGONOMETRY

x	0	$30°={}^{\pi}/_{6}$	$45°={}^{\pi}/_{4}$	$60°={}^{\pi}/_{3}$	$90°={}^{\pi}/_{2}$
$\sin(x)$	0	$^1/_2$	$^{\sqrt{2}}/_2$	$^{\sqrt{3}}/_2$	1
$\cos(x)$	1	$^{\sqrt{3}}/_2$	$^{\sqrt{2}}/_2$	$^1/_2$	0
$\tan(x)$	0	$^1/_{\sqrt{3}}$	1	$\sqrt{3}$	

$$\tan(x)=\frac{\sin(x)}{\cos(x)}$$
$$\sin(2x)=2\sin(x)\cos(x)$$
$$\sin^2(x)+\cos^2(x)=1$$

$1, \sin(\alpha), \cos(\alpha)$



VECTORS

Term	Definition
Norm/Length	$\lVert v\rVert=\sqrt{\sum_{i=1}^n v_i^2}=\sqrt{v^\top v}$ $\nabla\lVert v\rVert=\frac{1}{\lVert v\rVert}v^\top$ $\nabla\frac{1}{\lVert v\rVert}=-\frac{1}{\lVert v\rVert^2}\nabla\lVert v\rVert=-\frac{1}{\lVert v\rVert^3}v^\top$
Distance	$d(u,v)=\lVert u-v\rVert$ $\nabla d(u,v)=\frac{1}{d(u,v)}((u-v)^\top(v-u)^\top)$
Scalar product	$u\bullet v=u^\top v=\sum_k u_k v_k$ $\nabla(u\bullet v)=(v^\top\ u^\top)$

MATRICES

Affine transformation:

$$M\in\mathbb{R}^{m\times n},\quad A_{b,M}:\begin{cases}\mathbb{R}^m\to\mathbb{R}^n\\ x\ \ \mapsto b+M\cdot x\end{cases}$$

Determinant:

$$\det(a)=a \qquad \det\begin{pmatrix}a&b\\c&d\end{pmatrix}=ad-cb$$

$$\det\begin{pmatrix}a_{11}&a_{12}&a_{13}\\a_{21}&a_{22}&a_{23}\\a_{31}&a_{32}&a_{33}\end{pmatrix}=a_{11}a_{22}a_{33}+a_{12}a_{23}a_{31}+a_{13}a_{21}a_{32}$$
$$-a_{31}a_{22}a_{13}-a_{32}a_{23}a_{11}-a_{33}a_{21}a_{12}$$

$$\det(A\cdot B)=\det(A)\cdot\det(B) \qquad \det(\mathbb{1})=1$$
$$\det(A^{-1})=\frac{1}{\det(A)} \qquad \det(A^\top)=\det(A)$$

Inverting:

$$A=\begin{pmatrix}a&b\\c&d\end{pmatrix}\Rightarrow A^{-1}=\frac{1}{ad-cb}\begin{pmatrix}d&-b\\-c&a\end{pmatrix}$$

Transposing:

$$\begin{pmatrix}1&0\\0&2\\3&1\end{pmatrix}^\top=\begin{pmatrix}1&0&3\\0&2&1\end{pmatrix} \quad (A^\top)^\top=A \quad (A+B)^\top=A^\top+B^\top$$

$$\begin{pmatrix}1\\4\\5\end{pmatrix}^\top=(1\ 4\ 5) \quad (\lambda A)^\top=\lambda A^\top \quad (AB)^\top=B^\top A^\top$$

Matrix multiplication:

$$\begin{pmatrix}2&3&1\\4&-1&7\end{pmatrix}\cdot\begin{pmatrix}6&4\\1&0\\8&9\end{pmatrix}$$
$$=\begin{pmatrix}(2\cdot6+3\cdot1+1\cdot8)&(2\cdot4+3\cdot0+1\cdot9)\\(4\cdot6+-1\cdot1+7\cdot8)&(4\cdot4+-1\cdot0+7\cdot9)\end{pmatrix}=\begin{pmatrix}23&17\\79&79\end{pmatrix}$$

PROPERTIES

$$A\cdot A^{-1}=\mathbb{1} \qquad \ker A=\{v\mid Av=0\}$$
$$A+B=B+A \qquad A\cdot B\neq B\cdot A$$
$$A+(B+C)=(A+B)+C \qquad A(BC)=(AB)C$$
$$A(B+C)=AB+AC \qquad (A+B)C=AC+BC$$

INDICES

$$A\in\mathbb{R}^{n\times m}, B\in\mathbb{R}^{m\times r}, \qquad A\cdot B\in\mathbb{R}^{n\times r}$$
$$(A\cdot B)_{ij}=\sum_{k=1}^m A_{ik}B_{kj} \qquad =\sum_k A_{ik}B_{kj}$$
$$((A\cdot B)^\top)_{ij}=(A\cdot B)_{ji} \qquad =\sum_k A_{jk}B_{ki}$$
$$(A\cdot B^\top)_{ij}=\sum_k A_{ik}B^\top_{kj} \qquad =\sum_k A_{ik}B_{jk}$$
$$a^\top\cdot B\cdot b=\sum_{ij}(a^\top)_i B_{ij}b_j \qquad =\sum_{ij}a_iB_{ij}b_j$$
$$(B\cdot a)(C\cdot a)=\sum_i(B\cdot a)_i(C\cdot a)_i$$

ADDITION/SUBTRACTION

$$A\in\mathbb{R}^{n\times m}, B\in\mathbb{R}^{n\times m}$$
$$(A+B)_{ij}=A_{ij}+B_{ij}$$
$$(A-B)_{ij}=A_{ij}-B_{ij}$$

KRONECKER DELTA

$$\delta_{ij}=(\vec{e}_i)_j=\begin{cases}1\ , i=j\\0\ \ , \text{otherwise}\end{cases}$$
$$(E_{rst})_{ijk}=\delta_{ri}\delta_{sj}\delta_{tk}$$

BASIS VECTORS

$$\mathbb{R}^3$$
$$e_1=\begin{pmatrix}1\\0\\0\end{pmatrix}, e_2=\begin{pmatrix}0\\1\\0\end{pmatrix}, e_3=\begin{pmatrix}0\\0\\1\end{pmatrix}$$

UNIT MATRICES

$$\mathbb{R}^2$$
$$E_{11}=\begin{pmatrix}1&0\\0&0\end{pmatrix}, E_{12}=\begin{pmatrix}0&1\\0&0\end{pmatrix}, E_{21}=\begin{pmatrix}0&0\\1&0\end{pmatrix}, E_{22}=\begin{pmatrix}0&0\\0&1\end{pmatrix}$$

AFFINE TRANSFORMATIONS

$$A_{a,M}(x)=a+M\cdot x$$
$$L_A(\vec{0})=\vec{0}$$

DERIVATIVES

1	$\to 0$	x	$\to 1$
x^2	$\to 2x$	x^n	$\to nx^{n-1}$
$\frac{1}{x}$	$\to -\frac{1}{x^2}$	$\sqrt{x}$	$\to \frac{1}{2\sqrt{x}}$
e^x	$\to e^x$	e^{-x}	$\to -e^{-x}$
$\ln(x)$	$\to \frac{1}{x}$ for $x>0$	$\ln(y\cdot x)$	$\to \frac{1}{x}$ for $x>0$
a^x	$\to \ln(a)\cdot a^x$ for $a>0, a\neq 1$	$\log_b(x)$	$\to \frac{1}{\ln(b)\cdot x}$
$\sin(x)$	$\to \cos(x)$	$\sin(2x)$	$\to 2\cos(2x)$
$\cos(x)$	$\to -\sin(x)$	$\cos(ax)$	$\to -a\sin(ax)$
$\tan(x)$	$\to 1+\tan^2(x)=\frac{1}{\cos^2(x)}$	$\mathrm{arccot}(x)$	$\to -\frac{1}{1+x^2}$
$\cot(x)$	$\to -\frac{1}{\sin^2(x)}$	$\arcsin(x)$	$\to \frac{1}{\sqrt{1-x^2}}$
$\arccos(x)$	$\to -\frac{1}{\sqrt{1-x^2}}$	$\arctan(x)$	$\to \frac{1}{1+x^2}$

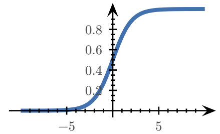
Term	Definition
Addition	$(f(x)+g(x))'=f'(x)+g'(x)$
Multiplication	$(\alpha\cdot f(x))'=\alpha\cdot f'(x)$
Product rule	$(f\cdot g)'=f'\cdot g+f\cdot g'$ $(f\cdot g\cdot h)'=f'\cdot g\cdot h+f\cdot g'\cdot h+f\cdot g\cdot h'$
Chain rule	$(f(g(x)))'=f'(g(x))\cdot g'(x)$
Quotient rule	$\left(\frac{f}{g}\right)'=\frac{f'g-fg'}{g^2}$

FUNCTIONS OF SEVERAL VARIABLES

Term	Definition
Real valued function	$R\subset\mathbb{R}$
Vector valued function	$R\subset\mathbb{R}^m$
A function of n variables	$D\subset\mathbb{R}^n$
n-dimensional curve	$c:\mathbb{R}\to\mathbb{R}^n$
n-dimensional hyper-surface	$s:\mathbb{R}^n\to\mathbb{R}$
Reparametrization	$c(t)\to d(s)=c(h(s))$

Sigmoid function:

$$\sigma(x):\mathbb{R}\to[0;1]=\frac{1}{1+e^{-x}}$$



DERIVATIVES

Function	Derivative
$f:\begin{cases}\mathbb{R}\to\mathbb{R}^n\\x\mapsto y\end{cases}$	$f':\begin{cases}\mathbb{R}\to\mathbb{R}^n\\x\mapsto y\end{cases}=\begin{pmatrix}\frac{\mathrm{d}}{\mathrm{d}x}f_1(x)\\\frac{\mathrm{d}}{\mathrm{d}x}f_2(x)\\\vdots\\\frac{\mathrm{d}}{\mathrm{d}x}f_n(x)\end{pmatrix}$
$f:\begin{cases}\mathbb{R}^n\to\mathbb{R}\\x\ \ \mapsto y\end{cases}$	$f':\begin{cases}\mathbb{R}^n\to\mathbb{R}^{1\times n}\\x\ \ \mapsto y\end{cases}=\nabla f$ $=\left(\frac{\partial}{\partial x_1}f(x),\frac{\partial}{\partial x_2}f(x),...,\frac{\partial}{\partial x_n}f(x)\right)$
$f:\begin{cases}\mathbb{R}^n\to\mathbb{R}^m\\x\ \ \mapsto y\end{cases}$	$f':\begin{cases}\mathbb{R}^n\to\mathbb{R}^{m\times n}\\x\ \ \mapsto y\end{cases}=J_f=\left.\frac{\partial f}{\partial x}\right\rvert_{x=x_0}$ $=\begin{pmatrix}\nabla f_1(x)\\\vdots\\\nabla f_m(x)\end{pmatrix}=\begin{pmatrix}\frac{\partial}{\partial x_1}f_1(x)&\dots&\frac{\partial}{\partial x_n}f_1(x)\\\vdots&\ddots&\vdots\\\frac{\partial}{\partial x_1}f_m(x)&\dots&\frac{\partial}{\partial x_n}f_m(x)\end{pmatrix}$

PARTIAL VS TOTAL DERIVATIVE

TODO:
example FS2024

$$f(x,y)=x^2+2y$$
$$\frac{\partial f}{\partial x}=2x$$
$$\frac{\mathrm{d}f}{\mathrm{d}x}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\cdot\frac{\mathrm{d}y}{\mathrm{d}x}=2x+2\cdot\frac{\mathrm{d}y}{\mathrm{d}x}\ \mid(\text{y dependent on x})$$

GRADIENT VS TOTAL DERIVATIVE

$$f(x,y)=xye^y$$
$$x(t)=e^t$$
$$y(t)=t^2$$
$$\nabla f=(ye^y,xe^y+xye^y) \qquad =e^y(y,x+xy)$$
$$\frac{\mathrm{d}}{\mathrm{d}t}f=e^y(y,x+xy)\cdot\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix}e^t\\t^2\end{pmatrix} \qquad =\frac{\mathrm{d}}{\mathrm{d}t}\left(e^tt^2e^{t^2}\right)$$
$$=(2t+t^2+2t^3)e^{t^2+t}$$

GRADIENTS

Common gradients:

$$f(x)=\lVert x\rVert^2 \qquad \Rightarrow\nabla f=2x$$
$$f(x)=a^\top x+c \qquad \Rightarrow\nabla f=a$$
$$f(x)=x^\top Ax \qquad \Rightarrow\nabla f=(A+A^\top)x$$
$$f(x)=\frac{1}{2}x^\top Ax+b^\top x+c\Rightarrow\nabla f=\frac{A+A^\top}{2}x+b$$
$$=Ax+b \text{ if } A^\top=A$$
$$f(x)=g(Ax+b) \qquad \Rightarrow\nabla f=A^\top\nabla g(Ax+b)$$
$$g:\mathbb{R}^m\to\mathbb{R}$$

JACOBIAN MATRIX

$$f:\mathbb{R}^n\to\mathbb{R}^m \qquad x_0\in\mathbb{R}^n$$
$$J_f(x_0):\mathbb{R}^{m\times n} \qquad J_f(x_0)_{ij}=\frac{\partial}{\partial x_j}f_i(x_0)$$
$$f:\mathbb{R}^2\to\mathbb{R}^2 \qquad J_f(x,y)=\begin{pmatrix}\partial x_1&\partial y_1\\\partial x_2&\partial y_2\end{pmatrix}$$

ty lukas <3

$$\begin{tikzcd}
\mathrm{Euclid}_* & & \mathrm{Mat}_{\mathbb{R}} \\
(\mathbb{R}^n,x_0) \arrow[d, "g"'] \arrow[dd, bend left, "f\circ g"] & & \mathbb{R}^n \arrow[d, "J_g(x_0)"'] \arrow[dd, bend left, "J_f(g(x_0))J_g(x_0)"] \\
(\mathbb{R}^m,g(x_0)) \arrow[d, "f"'] & \overset{D}{\Rightarrow} & \mathbb{R}^m \arrow[d, "J_f(g(x_0))"'] \\
(\mathbb{R}^k,f(g(x_0))) & & \mathbb{R}^k
\end{tikzcd}$$

Chain rule:

$$f:\mathbb{R}^n\to\mathbb{R}^m,\ g:\mathbb{R}^m\to\mathbb{R}^k,\ h:\mathbb{R}^n\to\mathbb{R}^k=g\circ f$$
$$J_h=J_g(f(x))\cdot J_f(x)$$

Common jacobians:

$$f(x)=a+M\cdot x\Rightarrow J_f=M$$
$$f(x)=x \qquad \Rightarrow J_f=\mathbb{1}$$
$$f(x)=\lVert x\rVert^2 \qquad \Rightarrow J_f=(2x)^\top$$

LINEARISATION

$$f:\mathbb{R}^n\to\mathbb{R}^m,\ x_0\in\mathbb{R}^n$$
$$L_f(x)=f(x_0)+J_f(x_0)\cdot(x-x_0)$$

For finding better approximations for x_0:
1) Calculate J_f, then $J_f(x_0)$, then L_f
2) Find x so that $L_f(x)\overset{!}{=}0$ (Gauss)

For calculating an approximate value at x:
1) x_0 = round x to ints
2) Calculate $L_f(x)$ at x_0

SECOND DERIVATIVE

$$\frac{\partial}{\partial x}\left(\frac{\partial}{\partial x}(f(x,y))\right)=\frac{\partial^2f(x,y)}{\partial x^2}=\partial_{xx}f(x,y)$$

HESSIAN MATRIX

$$f:\mathbb{R}^n\to\mathbb{R},\quad x\in\mathbb{R}^n,\quad H(x)\in\mathbb{R}^{n\times n}$$
$$H(x,y)=\begin{pmatrix}f_{xx}&f_{xy}\\f_{yx}&f_{yy}\end{pmatrix}=J(\nabla f)$$
$$(H(x))_{ij}=\frac{\partial}{\partial x_i}\frac{\partial}{\partial x_j}f(x)$$
$$\partial_i\partial_jf=\partial_j\partial_if$$

Common hessians:

$$f(x)=\frac{1}{2}x^\top Ax+b^\top x+c\Rightarrow H_f=\frac{A+A^\top}{2}$$
$$f(x)=g(Ax+b) \qquad \Rightarrow H_f=A^\top H_g(Ax+b)A$$
$$g:\mathbb{R}^m\to\mathbb{R}$$

EXTREMAL POINTS

Quadratic form:

$$M\in\mathbb{R}^{n\times n} \qquad q_M(v)=v^\top Mv \qquad v\in\mathbb{R}^n$$
$$H=\frac{1}{2}(M+M^\top) \quad \forall v\in\mathbb{R}^n.\ (q_M(v)=q_H(v))$$

$v^\top Hv>0\Rightarrow c_0$ is a ***local minimum***, H is ***positive definite***

$v^\top Hv\geq0\Rightarrow c_0$ is a ***local minimum*** or ***non-extremal***, H is ***positive semi-definite***

$v^\top Hv<0\Rightarrow c_0$ is a ***local maximum***, H is ***negative definite***

$v^\top Hv\leq0\Rightarrow c_0$ is a ***local maximum*** or ***non-extremal***, H is ***negative semi-definite***

$v^\top Hv\neq0\Rightarrow c_0$ is a ***saddle point***, H is ***indefinite***

Finding stationary points:
1) Calculate ∇f, then H_f
2) Find stationary point(s) x so that $\nabla f(x)\overset{!}{=}0$
3) Find type of point using technique of completing squares on $H_f(x)$

$$(v_1,v_2,v_3)\cdot\begin{pmatrix}h_{11}&h_{12}&h_{13}\\h_{21}&h_{22}&h_{23}\\h_{31}&h_{32}&h_{33}\end{pmatrix}\cdot\begin{pmatrix}v_1\\v_2\\v_3\end{pmatrix}$$
$$=(v_1,v_2,v_3)\begin{pmatrix}v_1h_{11}+v_2h_{12}+v_3h_{13}\\v_1h_{21}+v_2h_{22}+v_3h_{23}\\v_1h_{31}+v_2h_{32}+v_3h_{33}\end{pmatrix}$$
$$=v_1^2h_{11}+v_1v_2h_{12}+v_1v_3h_{13}$$
$$+v_1v_2h_{21}+v_2^2h_{22}+v_3v_2h_{23}$$
$$+v_1v_3h_{31}+v_2v_3h_{32}+v_3^2h_{33}$$
$$=v_1^2h_{11}+v_2^2h_{22}+v_3^2h_{33}+2v_1v_2h_{12}+2v_2v_3h_{23}+2v_1v_3h_{13}$$

Technique of completing squares:

$$\underbrace{x^2}_{\sqrt{\cdot}}-\underbrace{4xy}_{\div2x}+1$$
$$=(x-2y)^2-(-2y)^2+1$$
$$=(x-2y)^2-4y^2+1$$

HYPER-SURFACES

Chain rule:

$$f:\mathbb{R}^n\to\mathbb{R}^m,\ g:\mathbb{R}^m\to\mathbb{R},\ h:\mathbb{R}^n\to\mathbb{R}=g\circ f$$
$$\nabla h(x)=\nabla g(f(x))=\nabla g(f)|_{f=f(x)}\cdot\frac{\partial}{\partial x}f(x)$$

Differentiation properties:

$$f:\mathbb{R}^n\to\mathbb{R},\ g:\mathbb{R}^n\to\mathbb{R}$$
$$\nabla(f+g)=\nabla f+\nabla g$$
$$\nabla(f-g)=\nabla f-\nabla g$$
$$\nabla(f\cdot g)=g\nabla f+f\nabla g$$
$$\nabla\frac{f}{g}=\frac{g\nabla f-f\nabla g}{g^2}$$

COMMUTATIVE DIAGRAMS

$f:D\to M$
$g:N\to R$
$h:D\to R$
$\ \ =g\circ f$

$$\begin{tikzcd}
D \arrow[r, "f"] \arrow[rr, bend left, "g\circ f"] & M \arrow[r, "g"] & R
\end{tikzcd}$$

GRADIENT DESCENT / NEWTON'S METHOD

$$GD \qquad x_{i+1}=x_i-\gamma\cdot\nabla f(x_i)$$
$$N \qquad x_{i+1}=x_i-\gamma\cdot\left(H_{f(x_i)}\right)^{-1}\cdot\nabla f(x_i)$$
$$\gamma=\text{step size}$$
$$\text{termination criterion } \varepsilon>|x_{i+1}-x_i|$$

PROBABILITY THEORY

Term	Definition
$\sim$	Distributed
$\mathcal{N}(\mu,\sigma)$	Normal distribution
$\mathrm{unif}(a,b)$	Uniform distribution
$\mathbb{P}(E)$	Probability measure
$l(\mu,\sigma)$	Likelihood function $\prod_{i=1}^n f(x_i\mid\mu,\sigma)$
$\ln(l(\mu,\sigma))$	Log-likelihood function $\sum_{i=1}^n\ln(f(x_i\mid\mu,\sigma))$
CDF	Cumulative Distribution Function F
PDF	Probability Density Function f

TODO:

$$\text{density} = f(x_i \mid \mu, \sigma)$$

KOLMOGOROV AXIOMS

$$\mathbb{P}(\Omega) = 1$$
$$\mathbb{P}(\varnothing) = 0$$
$$\forall A, B \subset \Omega,\ A \cap B = \varnothing.\ (\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B))$$
$$\forall A, B \subset \Omega.\ (\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B))$$

STATISTICALLY INDEPENDENT

$X : \Omega \to \mathbb{R}$ and $Y : \Omega \to \mathbb{R}$ are ***statistically independent***, if the probability density f_V of the random variable

$$V : \begin{cases} \Omega \to \mathbb{R}^2 \\ \omega \mapsto \binom{X(\omega)}{Y(\omega)} \end{cases}$$

satisfies

$$f_V(x, y) = f_X(x) \cdot f_Y(y)$$

DISCRETE PROBABILITY DISTRIBUTION

Let $\Omega = \{e_1, e_2, ..., e_n\}$ be enumerable

$$\mathbb{P}(\{e_k\}) = p_k, \quad k \in \{1, 2, ..., n\}, \quad E \subset \Omega$$
$$\mathbb{P}(E) = \sum_{e \in E} p(e)$$
$$F(x) = \sum_{e \in \Omega, e \leq x} p(e)$$

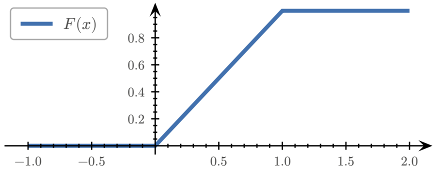
RULES

$$\forall e \in \Omega.\ (0 \leq p(e) \leq 1)$$

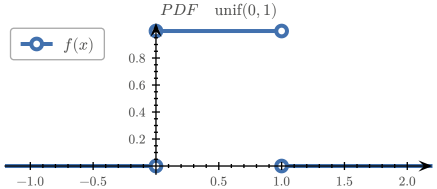
$$\sum_{e \in \Omega} p(e) = 1 \qquad \int_{-\infty}^{\infty} f(t)\,\mathrm{d}t = 1$$
$$\lim_{x \to -\infty} F(x) = 0 \qquad \lim_{x \to \infty} F(x) = 1$$
$$0 \leq F(x) \leq 1 \qquad f(t) \geq 0 \text{ for } t \in \mathbb{R}$$

$F(x)$ is monotonically increasing & continuous from the right

$CDF \quad \mathrm{unif}(0,1)$



$PDF \quad \mathrm{unif}(0,1)$



UNIVARIATE UNIFORM DISTRIBUTION

$$X \sim \mathrm{unif}(a, b), \quad \Omega \subset \mathbb{R}$$

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{if } x \in (a;b) \\ 0 & \text{else} \end{cases} = \mathbb{1}_{(a;b)}(x) \cdot \frac{1}{b-a} = F_X'$$

$$F_X(x) = \mathbb{P}((-\infty; x] \cap \Omega) \quad = \mathbb{1}_{[a;b]}(x) \cdot \frac{x-a}{b-a} + \mathbb{1}_{(b;\infty)}(x)$$
$$= \mathbb{P}(X \leq x) \quad = \int_{-\infty}^{x} f_X(t)\,\mathrm{d}t$$

UNIVARIATE NORMAL DISTRIBUTION

Standard normal distribution:

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

General normal distribution:

$$f(x|\mu, \sigma) = \frac{1}{\sigma} \varphi\left(\frac{x-\mu}{\sigma}\right)$$

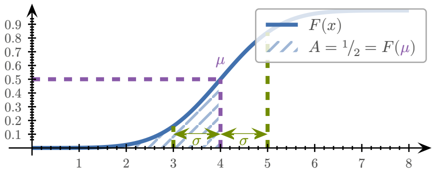
Univariate normal distribution:

$$x \sim \mathcal{N}(\mu, \sigma)$$
$$f(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$
$$\mu \approx \overline{x} = \frac{1}{N} \sum_{i=1}^{N} x_i$$
$$\sigma^2 \approx \mathrm{var} = \frac{1}{N-1} \sum_{i=1}^{N} (x_i - \overline{x})^2$$

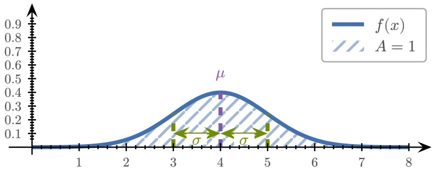
Error function:

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\,\mathrm{d}t$$

CDF



PDF



mean value (μ), standard deviation (σ)

MULTIVARIATE NORMAL DISTRIBUTION

$$\Sigma = \text{covariance matrix}, \quad V : \Omega \to \mathbb{R}^n$$
$$V \sim \mathcal{N}(\mu, \Sigma)$$
$$f_V(v) = \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} \cdot e^{q_{\Sigma^{-1}}(v-\mu)}$$
$$q_{\Sigma^{-1}}(v - \mu) = -\frac{1}{2}(v-\mu)^\top \Sigma^{-1} (v - \mu)$$

Sample mean:

$$\mu \approx \overline{v} = \frac{1}{N} \sum_{i=1}^{N} v_i$$

Sample covariance matrix:

$$\Sigma \approx Q = \frac{1}{N-1} \sum_{i=1}^{N} (v_i - \overline{v})(v_i - \overline{v})^\top$$

EXAMPLES

FINDING LIKELIHOOD FUNCTION

Given the following data and the probability density of the form

x	y
1	7
2	16
3	32
4	44

$$f(t) = \lambda e^{-\lambda t} \cdot 1_{(0;\infty)}(t)$$

find a formula for the likelihood function

$$l(\lambda) = f(7) \cdot f(16) \cdot f(32) \cdot f(44) = \lambda^4 e^{-99\lambda}$$

Find a formula for the negative log-likelihood

$$-\ln(l(\lambda)) = -\ln\left(\lambda^4 e^{-99\lambda}\right) = 99\lambda - 4\ln(\lambda)$$

TODO:
- calculate probability density function
- gradient descent
- check if function is increasing ($f' > 0$) → Aufgabe 123/124

- distribution to density and vice-versa $X = g(Y) \wedge g$ increasing $\Rightarrow f_Y(y) = f_X(g(y)) \cdot g'(y)$
- FS2024 7.b)
- FS2024 5.b)
- FS2024 6.a)b)

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} x_i$$
$$\hat{\sigma} = \left(\frac{1}{n} \sum_{i=1}^{n} (x_i - \mu)^2\right)^{\frac{1}{2}}$$

WORKING WITH INDICES

$$x, y \in \mathbb{R}^n, \ f : \mathbb{R}^n \to \mathbb{R}$$
$$f(x) = (x-y)^\top (3x - y) = \sum_{i=1}^{n} (x_i - y_i)(3x_i - y_i)$$
$$\partial x_i f(x) = (3x_i - y_i) + 3(x_i - y_i)$$
$$f(x) = \ln(x_1 \cdot x_2 \cdot ... \cdot x_n) - |x|^2 = \sum_{i=1}^{n} (\ln(x_i) - x_i^2)$$
$$\partial x_i f(x) = \frac{1}{x_i} - 2x_i$$
$$f(x) = \ln(y^\top x) = \ln\left(\sum_{i=1}^{n} y_i x_i\right) \Rightarrow \nabla f = \frac{1}{y^\top x} y^\top$$